

Detecting Urban Food Deserts Using Spatial Co-location Mining

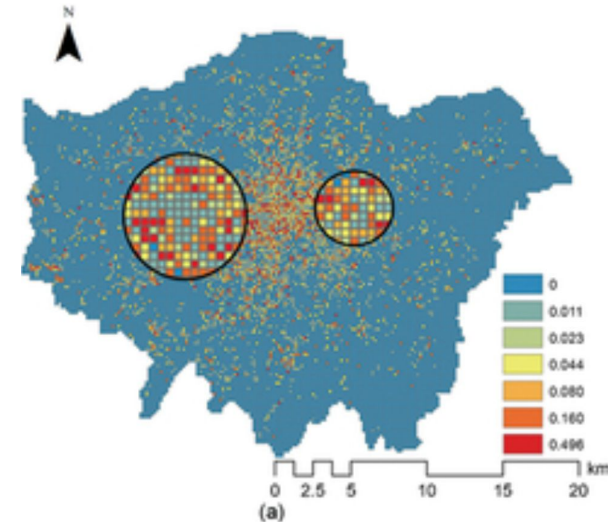
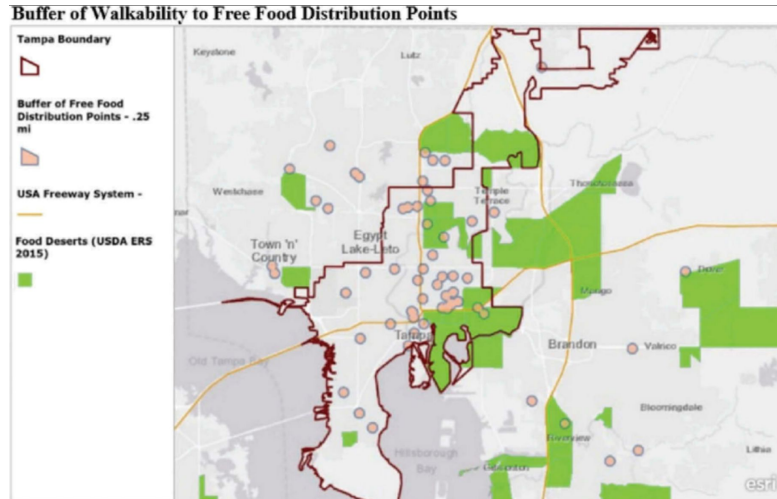
Woochang Shin and Yashasvi Singh



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Research Question

How can we detect true food deserts using **spatial co-location patterns** rather than simple **distance-based** definitions?



Distance-Based Definition (Buffer Model) vs Spatial Co-Location Pattern



Societal Motivation

- Food access inequality is a major urban challenge
- Limited access to healthy and affordable food is linked to obesity, diabetes, and chronic disease
- Low-income neighborhoods often experience systematic grocery underrepresentation
- Accurate detection of food environment imbalance can improve public policy and community health

DAILY NEWS CLIP: JUNE 30, 2025

Nearly 19 million people in the US live in a food desert: Here's what that means

Hartford Courant - Saturday, June 28, 2025

By Hadley Mendelsohn

Nearly 19 million people in the US live in a food desert: Here's what that means

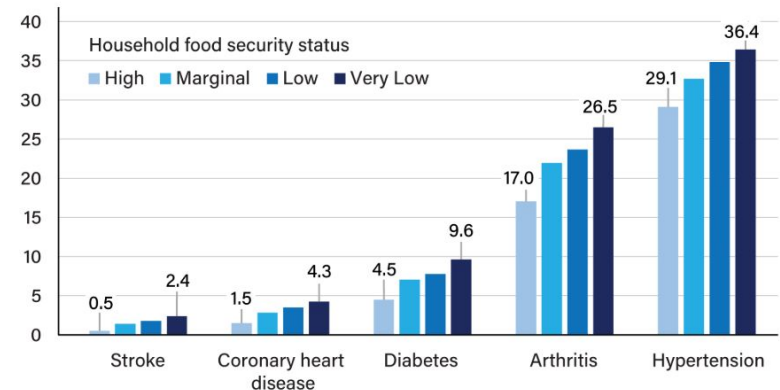
We talk a lot about eating healthier — more greens, less sugar, fewer processed foods. But what if access to those healthier choices isn't actually a choice for everyone?

Obesity is an increasingly common health issue in the U.S., and it's linked to higher

Predicted disease prevalences for adults in low-income households, 2019–22

USDA Economic Research Service
U.S. DEPARTMENT OF AGRICULTURE

Predicted prevalence (percent)



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Problem Statement

Data Components:

- Geolocated food outlet data (type, latitude, longitude, etc.)
- Neighborhood definition
- Interaction radius R

Expected Outcomes:

- Identification of structurally imbalanced neighborhoods
- Co-location interaction metrics
- Comparison with distance-based classification

Study Goal:

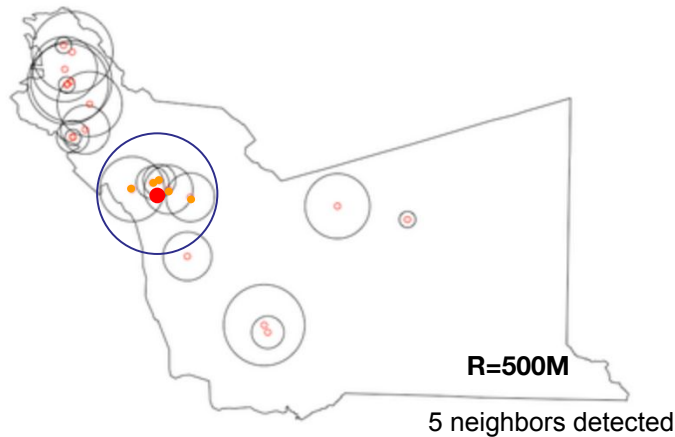
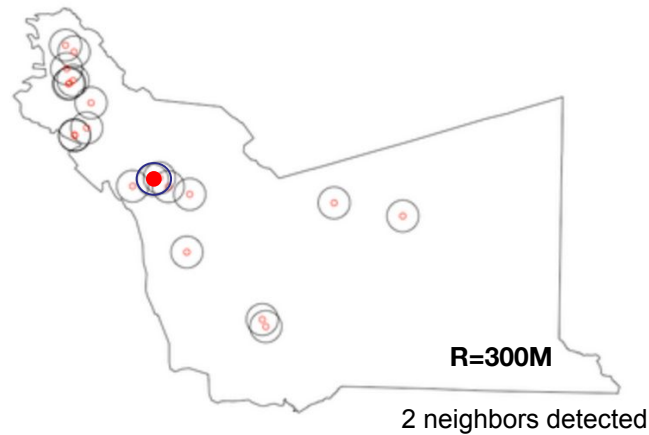
- Detect spatial imbalance in food environments

Constraints:

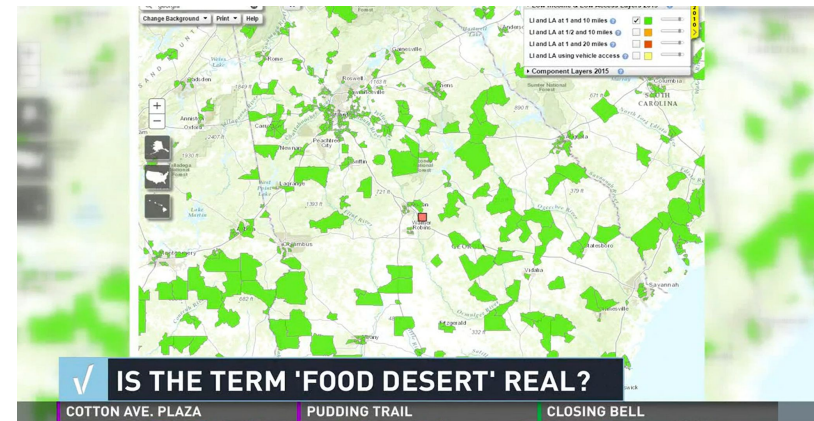
- Continuous spatial data (no natural transactions)
- Sensitivity to neighborhood radius R
- Uneven urban density



Key Concepts



Spatial Neighborhood ®



Food Desert

$$PR(C, f_i) = \frac{|Instances(f_i \text{ in } C)|}{|Instances(f_i)|}$$

Formula: Participation Ratio Formula

$$PI(C) = \min_{f_i \in C} PR(C, f_i)$$

Formula: Participation Index

Participation Index (PI)



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Challenges

No natural “transactions” in continuous spatial data (need neighborhood definition)

Neighborhood radius selection (R) affects results scale sensitivity

Data sparsity & imbalance across neighborhoods (downtown vs residential)

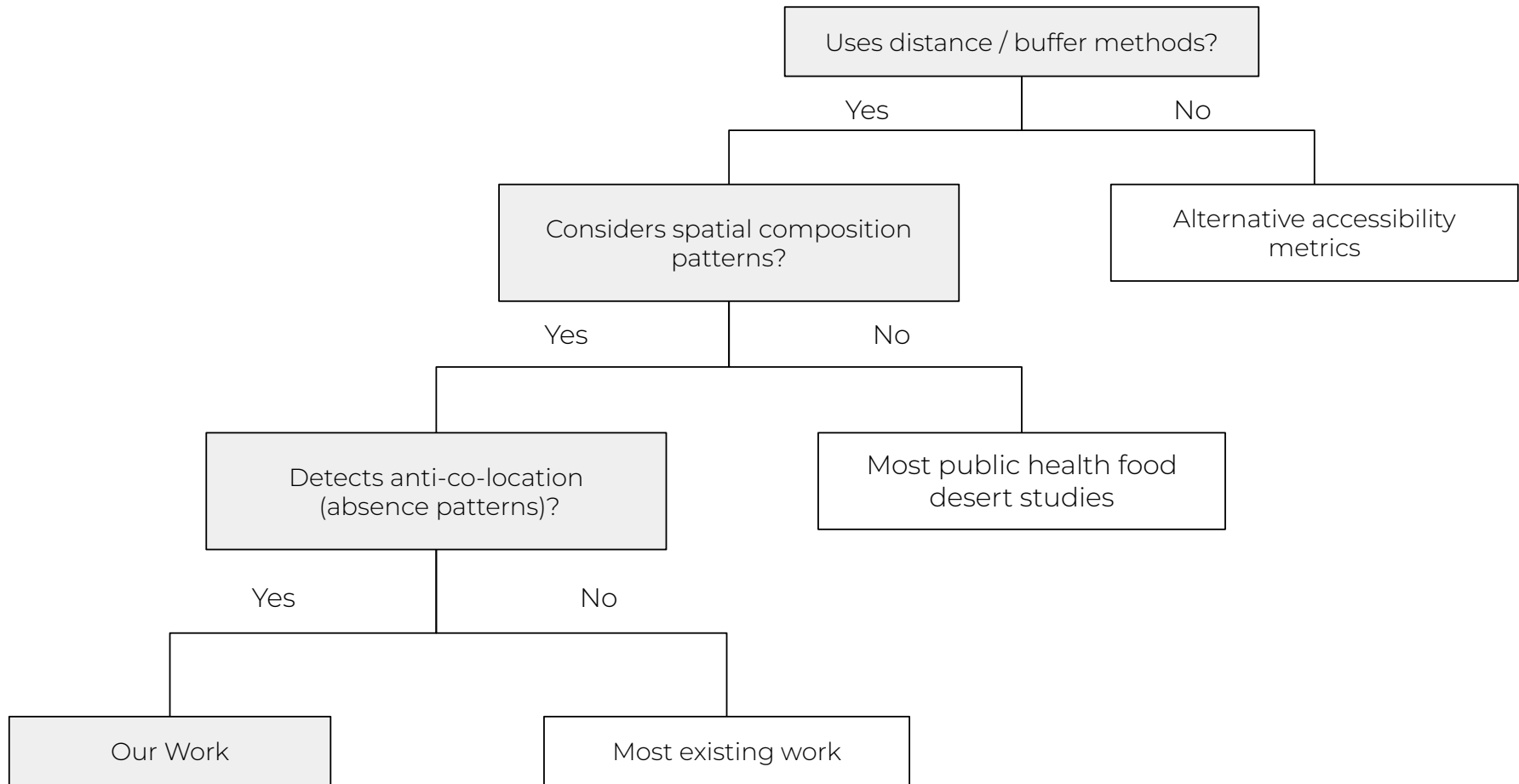
Capturing absence / *anti-co-location (e.g., fast-food present but grocery missing)

*Anti-co-location is the spatial association between presence of one feature and absence of another.



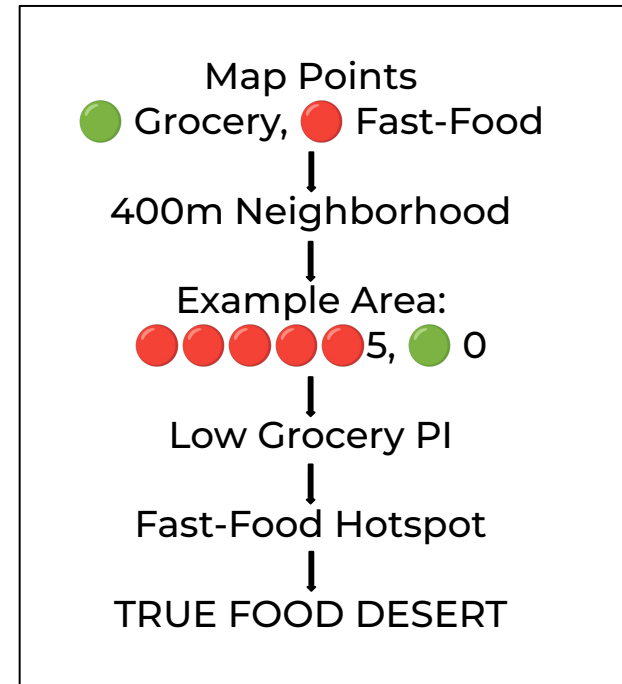
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Related Work and Its Limitations



Proposed Approach

- Data Representation
- Spatial Neighborhood Construction
- Co-location Pattern Mining
- Anti-Co-location Detection
- Hotspot Integration
- Spatial Classification Framework



Validation Methodology

Controlled Example (Synthetic)

- Create simple artificial regions
- Design 3 patterns:
 - Balanced
 - Fast-food dominant
 - Grocery absence
- Verify correct classification

Real Data Experiment

- Apply to Minneapolis (or other city) POI data
- Test radius (300m / 400m / 500m)
- Compare with proximity-based method
- Check result stability
- Baseline: 1-mile supermarket buffer model
- Evaluation metrics: Precision / Recall
- Sensitivity analysis on PI threshold



Conclusions and Future Work

Conclusions

- Spatial co-location framework for food desert detection
- Combines hotspot analysis + PI-based mining
- Produces interpretable urban service classification

Future Work

- Add more POI categories
- Integrate socio-economic layers
- Multi-city / temporal extension



Timeline

Week	Tasks
March 3 - Mar 8	• Research design & literature review • Data collection
March 16 - Mar 23	• Data preprocessing • Framework design
Mar 24 - Mar 30	• Implement PI computation • Develop classification rules
Mar 31 - April 6	• Apply to real data • Validation & baseline comparison
April 7 - April 16	• Result analysis • Prepare maps & presentation materials



Questions?



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Thank you

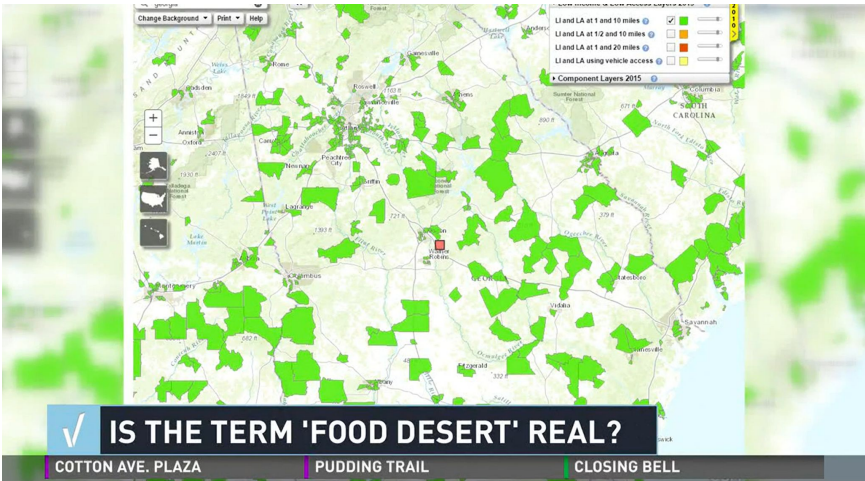


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Key Concepts

Food Desert

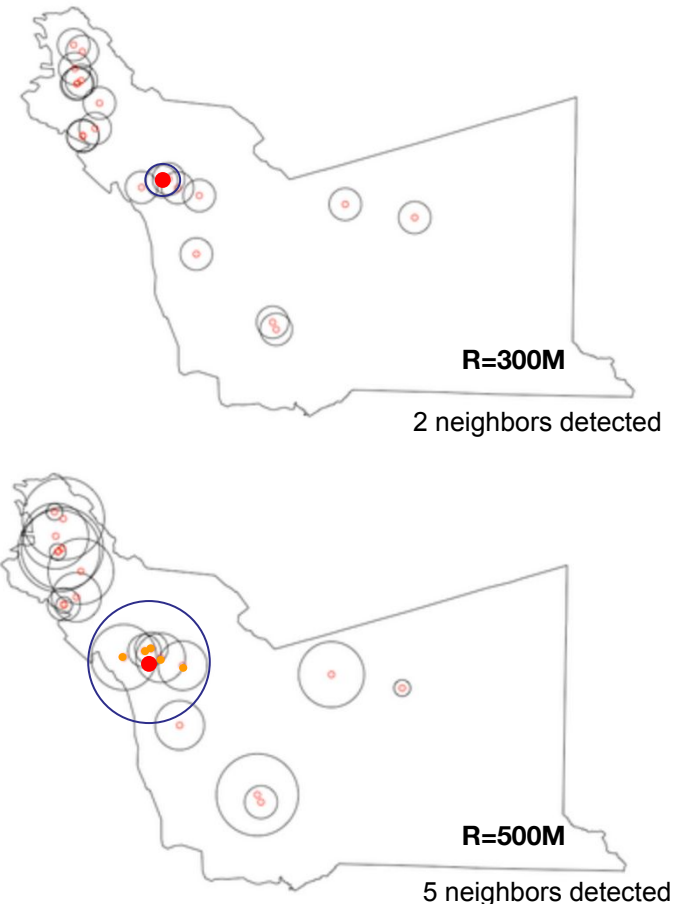
- Areas with limited access to affordable and nutritious grocery options.
- Often characterized by a high density of fast-food or convenience stores relative to grocery stores.
- Reflects structural imbalance in local food environments rather than simple distance measures.
 - Not just proximity, but spatial composition matters.



Key Concepts

Spatial Neighborhood (R)

- A predefined geographic radius used to define local spatial interactions.
- Determines which food outlets are considered “neighbors.”
- Changing R can alter detected spatial patterns (sensitivity issue).
 - Spatial results depend on neighborhood scale.



Key Concepts

Participation Index (PI)

- Computed as the minimum participation ratio across all event types in a pattern.
- Measures the strength and prevalence of a co-location pattern.
- A high PI indicates strong spatial coupling between event types.
- Ensures balanced participation among categories.

$$PR(C, f_i) = \frac{|Instances(f_i \text{ in } C)|}{|Instances(f_i)|}$$

Formula: Participation Ratio Formula

$$PI(C) = \min_{f_i \in C} PR(C, f_i)$$

Formula: Participation Index

Example: C = {Grocery, FastFood}

Total Grocery stores: 10

Participating Grocery stores: 6

$PR(C, \text{Grocery}) = 6 / 10 = 0.6$

Total Fast-Food outlets: 8

Participating Fast-Food outlets: 4

$PR(C, \text{FastFood}) = 4 / 8 = 0.5$

$PI(C) = \min(0.6, 0.5) = 0.5$

